

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_7049e1ed-b93\agent-af134ff66471ced66.jsonl`  
- SHA-256: `6547b5ef67954938d9a0847a615222a831d2ac2d3260e9d9baa48dad01548522`  
- Source modified: `2026-07-02T04:53:09+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_7049e1ed-b93`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T04:45:39.420Z)

You are a senior engineer producing ONE isolated PR. Today is 2026-07-02.

REPO: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer (a shared checkout ? follow the safety rules exactly).

AUDIT FINDINGS YOU ARE FIXING: R1-19 (upload_dir CWD-relative) ? Grep

C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/AUDIT_REPORT_khelsutra-guru_2026-07-02.md for these IDs and read the full evidence + verifier notes before coding.

TASK BRIEF:

In C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer: backend/api/uploads.py:50 ? upload_dir default 'output/uploads' is CWD-relative while DB/reels are app-home-rooted. Resolve upload_dir through the same app-home resolution as output_dir (find resolve_output_dir/app_path in backend/utils/paths.py or config; relative ? app-home-rooted; absolute passes through), mirroring output_dir semantics exactly. Check the packaged-app path (install.py:243 advertises 'indexer --app' from any shell CWD). Tests: relative default resolves under app home; absolute honored; existing tests updated honestly (if a test pinned the CWD-relative behavior, change it and say so in the PR body).

BRANCHING SAFETY (mandatory ? shared checkout, sibling sessions exist):

1. In the MAIN repo checkout run: git fetch origin
2. Create an ISOLATED WORKTREE: git worktree add "C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-<shortname>" -b <branch-name> origin/master (branch explicitly FROM origin/master ? never from the current local branch; use origin/main for repos whose default is main)
3. Do ALL work inside the worktree. Before committing: git branch --show-current + git log --oneline -1 must show YOUR branch on the origin-master base.
4. Stage EXPLICIT paths only (git add <file> <file>) ? NEVER git add -A / . / -u
5. Before opening the PR: git log --oneline origin/master..HEAD must list ONLY your commits.

COMMIT/PR CONVENTIONS:

- Commit message: conventional style matching the repo's history; end the message with a blank line then: Co-Authored-By: Claude Fable 5 <noreply@anthropic.com>
- PR body: what/why, a "## Verification" section stating exactly what you ran locally and what CI covers, and end with: ? Generated with [Claude Code](https://claude.com/claude-code)
- Do NOT merge the PR. Do NOT remove the worktree (a fixer may need it). Return the PR number, branch, worktree path.

QUALITY BAR:

- Read the repo's CLAUDE.md first; for the engine also docs/CODE_MAP.md \$0 (file placement) before adding any new file.
- Add/extend tests for every behavior change. Run the targeted tests you touched (python -m pytest tests/<relevant> -q) + ruff check on changed files; if mypy config exists, run mypy on touched modules. Full suite is CI's job ? do not run it.
- Default-safe: new knobs default OFF / behavior-preserving unless the task explicitly says otherwise.
- The full audit evidence is in C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-

Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/AUDIT_REPORT_khelsutra-guru_2026-07-02.md ? Grep it for your finding IDs and READ the verifier notes; they contain corrections (exact file:line, scoping traps) you MUST honor.

If the brief's premise turns out false against current code (something already fixed, file moved), adapt honestly and say so in notes ? never force a pointless diff. If you genuinely cannot complete (env broken, auth missing), open NO PR and explain in notes with pr_number=0.

2. user (2026-07-02T04:45:46.811Z)

```
184-[Omitted long context line]
185-[Omitted long context line]
186-
187-### R1-16 [P2/L] tech-debt ? badminton-highlight-indexer
188-**Decision-snapshot regression track: decisions locked 06-25, zero code shipped since;
tracker P0 row also stale**
189-- anchor: `docs/DECISION_SNAPSHOT_REGRESSION.md:66` · tracked in:
docs/DECISION_SNAPSHOT_REGRESSION.md §7 (P1-P8) · finder: engine:audit-doc-reconcile
190-- evidence: §7 tracker (lines 63-73): P1-P8 all ?. Verified in code: grep
'build_stitch_plan' across backend/ and tests/ returns nothing (P2's pure stitch-plan seam was
never factored out of backend/pipeline/stitcher/compiler.py), and tests/test_golden_fixtures.py
has no stitch-plan snapshot. Minor rot: the P0 row (line 65) still shows ? '(this PR)' though
the doc merged as #383 (commit 1283e18, 2026-06-25) and the header still says 'DRAFT'.
191-- why it matters: This gate exists precisely to bless genuine (non-AST-identical) refactors
as quality-neutral ? i.e., exactly the class of change as the stalled #390 main.py split.
Executing P1-P4 first would de-risk redoing the god-module split; leaving it a week-stale
locked-decisions doc means refactors continue with no behavioral proof.
192-- proposed fix: Start §7 P1+P2 (canonical decision-snapshot schema + factor pure
build_stitch_plan() out of compiler.py) as the next Claude-doable engine work; flip the P0 row
to ? #383 and status to IN PROGRESS in the same PR. Ideally land P1-P4 before re-cutting the
main.py split.
193-[Omitted long context line]
194-[Omitted long context line]
195-
196-### R1-17 [P2/M] tech-debt ? badminton-highlight-indexer
197-**Fetched-input scratch copies are never reclaimed anywhere (process jobs, compile jobs,
/api/assets, cloud_run marker reads) ? 'OS temp reclamation' premise is false on Windows**
198-- anchor: `backend/storage/__init__.py:89` · tracked in: ? · finder: engine:correctness
199-[Omitted long context line]
200-- why it matters: Slow-burn disk exhaustion on exactly the appliance the product ships as;
also inflates the free-space guard's false rejections.
201-- proposed fix: Wrap remote-input acquisition in a per-job scope that deletes the scratch
dir in a finally (worker already owns the job lifecycle), or route scratch under
output_base(cfg)/tmp with an age-based sweep at startup; make cloud_run._read_json use
TemporaryDirectory.
202-[Omitted long context line]
203-[Omitted long context line]
204-
205-### R1-18 [P2/S] critical-fix ? badminton-highlight-indexer
206-**GDriveApiStorage.fetch_to_local buffers the entire file in RAM (get_media().execute());
put() deletes the old object before the new upload succeeds**
207-- anchor: `backend/storage/gdrive_api.py:144` · tracked in: ? · finder: engine:correctness
208-- evidence: gdrive_api.py:143-148: `data =
self._service.files().get_media(fileId=file_id).execute()` then `fh.write(data)` ? the whole
object is materialized as one bytes value (no MediaIoBaseDownload streaming).
gdrive_api.py:163-170: put() resolves an existing file at the path and `files().delete(...)`
BEFORE `files().create(...)` ? delete-then-create is not atomic.
209-- why it matters: Latent OOM + data-loss on the Drive-API storage hot path before it goes
live for the reel round-trip.
210-- proposed fix: Stream downloads with googleapiclient.http.MediaIoBaseDownload in chunks;
for put(), create the new file first (or upload to a temp name) and delete the old one only
after success.
211-[Omitted long context line]
212-[Omitted long context line]
```

213-

214-### R1-19 [P2/S] critical-fix ? badminton-highlight-indexer

215-****config.json output.output_dir is silently clobbered by the CLI --output-dir DEFAULT on every server/CLI start ? the typed knob is dead, and the DB path moves with it****

216-- anchor: `backend/main.py:1992` · tracked in: ? · finder: engine:correctness

217-- evidence: main.py:1837-1841 defines `--output-dir` with default='output'; main.py:1992 unconditionally does `global\_config.output.output\_dir = resolve\_output\_dir(args.output\_dir)` ? the override fires even when the flag was never passed, so a config.json value can never take effect. models.py:274-276 documents the knob as if it's honored ('The CLI's --output-dir overrides this at startup' ? implying only when given). db_path derives from it (main.py:1995-1997).

218-- why it matters: A documented config knob that is a no-op on the main entry path, with data-location consequences (perceived data loss / split state).

219-- proposed fix: Change argparse default to None and only override when args.output_dir is not None (falling back to the config value through resolve_output_dir).

220-[Omitted long context line]

221-[Omitted long context line]

222-

223-### R1-20 [P2/S] critical-fix ? badminton-highlight-indexer

224-****upload_dir default 'output/uploads' is CWD-relative and never app-home-resolved ? packaged --app writes uploads into an arbitrary CWD (or 500s on an unwritable one) while DB/reels live under RALLY_APP_HOME****

225-- anchor: `backend/api/uploads.py:50` · tracked in: ? · finder: engine:correctness

226-- evidence: uploads.py:50 `upload\_dir = getattr(sec, 'upload\_dir', None) or 'output/uploads'` used verbatim at :89 (os.makedirs), :95, :170 ? grep confirms no resolve_output_dir/app_path is ever applied to it (only output.output_dir is resolved, main.py:1992 via app_home.py:83-100). models.py:336 default 'output/uploads'.

227-- why it matters: Breaks or misplaces the primary ingest path of the packaged app; exactly the class of CWD-vs-app-home bug P0a was built to eliminate (app_home.py:1-8).

228-- proposed fix: Resolve upload_dir through resolve_output_dir()/app_path() in _upload_cfg (relative ? app-home-rooted; absolute passes through), mirroring output_dir.

229-[Omitted long context line]

230-[Omitted long context line]

231-

232-### R1-21 [P2/M] critical-fix ? badminton-highlight-indexer

233-****Cloud Run dispatch: a bucket-poll timeout abandons a still-running GPU execution ? no cancellation, so the L4 keeps billing and its late result is never ingested****

234-- anchor: `backend/compute/cloud\_run.py:173` · tracked in: ? · finder: engine:correctness

235-- evidence: cloud_run.py:173-178: run_infer blocks on poll_until(...); execution_policy.py:183-189 raises PolicyTimeout after max_wait_sec (default 1800s). The exception propagates to main.py:1068-1070 which marks the job failed (`cloud dispatch error`) ? no code anywhere in cloud_run.py calls executions.cancel/delete (grep: no cancel API), and the late-arriving done-marker/intervals.csv are never ingested (the local job is already terminal).

236-- why it matters: Direct GPU cost leak plus wasted results on the cloud-serving path (owner-gated go-live area, but the code path is live behind compute='cloud').

237-- proposed fix: On PolicyTimeout, best-effort cancel the execution via the Jobs API (executions.cancel with the stored execution name) and log the execution id in the failure message; optionally check the done-marker once more before failing.

238-[Omitted long context line]

239-[Omitted long context line]

240-

241-### R1-22 [P2/M] tech-debt ? badminton-highlight-indexer

242-****Stage-2 AI-handoff orchestration duplicated between the default segmenter (yolo_hybrid) and the trajectory family****

243-- anchor: `backend/pipeline/segmenters/trajectory\_hybrid.py:390` · tracked in: ? · finder: engine:structure

244-[Omitted long context line]

3. user (2026-07-02T04:45:47.269Z)

```

1      # CLAUDE.md ? project memory / agent handoff
2
3      Local, AI-powered **badminton (? racket-sports) highlight indexer + rally detector**:
4      FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid **Gemini** as the cloud AI. Sibling
5      repos: `sports-data-collector`, a VLC-based `rally-annotator`.

```

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6
7     ## Read these first
8     - **`docs/README.md`** ? doc index. **`docs/CODE_MAP.md`** ? code navigation + data
9       contracts (the cold-start map; read before touching code).
10    - **Authoritative owned-model roadmap:** `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+
the
11      live `docs/NEXT_STEPS.md`). These SUPERSEDE the 2026-06-07 strategy docs below where
they
12      disagree (licensing, Gemma-4 stance, base model, conversational-QA).
13    - **Strategy (post-scaffolding):** `docs/COMMERCIALIZATION.md`,
`docs/PMF_MARKET_RESEARCH.md`,
14      `docs/PLATFORM_ARCHITECTURE.md`, `docs/DEFERRED_HARDENING_PLAN.md`. Also
`docs/ROADMAP.md`
15      (offline-segmenter narrative) and `docs/BACKLOG.md`.
16    - **Rally-detection quality track (design merged 2026-06-13, owner-gated, implementation
in progress):**
17      `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model) ?
`docs/MULTI_SIGNAL_FUSION_PLAN.md`
18      (fuse shuttle + person + audio cues; the held-shuttle bug is the already-built
`rally_gate.py`
19      "Gate A" **not wired into production** ? cheapest win) ? `docs/EXPERIMENT_HARNESS.md`
(A/B + shadow
20      eval so cues land flag-gated/default-OFF with zero regression) ?
`docs/ROORA_LABELING_GUIDELINES.md`
21      (v8 vendor labeling spec). **`docs/NEXT_STEPS.md` has the prioritized pick-up for this
track.**
22    - **History:** `docs/archives/` ? ADRs (`decisions/DECISIONS.md`), research, and
23      **`checkpoints/`** (latest: `2026-06-strategy-foundation/`). **Archive policy:** only
move
24      **terminal-state** (DONE/REJECTED) work; live docs stay at `docs/` top level ? and
**before
25      archiving, HONESTLY update the doc first** (verify claims vs code ? flip status + a
completion
26      note ? update inbound refs ? keep the lineage ? then `git mv`), per the checklist +
living
27      lifecycle register in **`docs/DOC_STATUS.md`**.
28
29    ## Current state (June 2026)
30    - **Scaffolding complete** ; strategy foundation + owned-model roadmap merged. **No
31      platform/owned-model implementation has started.** A 2026-06-10 audit-driven hardening
32      sweep landed (licensing gate, detector keying, re-ingest safety, config, eval gate,
API
33      validation, reliability, CI + test coverage) ? see git history.
34    - **Two tracks, both owner-gated:** the **committed next PLATFORM slice = async job
model**
35      (`DEFERRED_HARDENING_PLAN.md` #16, single-tenant); the **owned-model track** starts at
a
36      greenlit Phase-0 milestone (`OWNED_MODEL_IMPLEMENTATION_PLAN.md`). **Do not start
37      implementation without explicit owner approval.**
38
39    ## Architecture in one breath
40    - **Pluggable registries** (ABC + `Registry` singleton): segmenters / providers /
validators
41      / sports / annotations ? see `backend/pipeline/segmenters/base.py`. *Future:*
42      `StorageBackend` + `ComputeBackend` registries (PLATFORM §3).
43    - `video_id` = MD5 of the file ? `backend/utils/hashing.py::compute_video_id` (one
helper).
44    - Typed config = `backend/config/` (Pydantic). DB schema evolves via the `PRAGMA
45      user_version` migration ladder in `backend/infrastructure/database.py`.
46
47    ## Committed guardrails (non-negotiable)
48    - **Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
49      training data source** (terms/copyright). Commercial-clean licensing for EVERY
dependency ?
50      code, data, AND weights: **MIT / Apache-2.0 / BSD only** (ratified 2026-06-09, owner-

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```

agreed).
51     - **Owned-model base:** **Qwen3-VL (Apache-2.0)** primary, InternVL3 (MIT) fallback;
**Gemma-4
52     is AMBER / bench-only** (Apache grant likely but Prohibited-Use posture unconfirmed ?
counsel
53     needed; do NOT build on it without sign-off). **Reject Gemma 3** (viral license).
**Exclude
54     minors** from the training pool; per-user training data needs a separate explicit opt-
in.
55     - **v1 = rally + highlight + history + correction loop**; **gait/biometrics strictly
future**
56     (GDPR Art. 9).
57     - **Generic data schema** ; `coach ? athlete` is an **optional overlay** , not baked in.
58     - **Compute = GCP ONLY (ADR-013, 2026-06-20):** GPU training + serving run on **GCP**
(`deploy/gcp/` ;
59     Cloud Run+GPU for serving per ADR-012). **DigitalOcean is DROPPED for compute** (DO
GPU not enabled
60     on the account + not cleanly credit-funded; Paperspace is subscription-gated). Don't
re-evaluate
61     DO/Paperspace for compute without a new ADR; the `$200` DO credits go to **non-compute
only** . The
62     `deploy/digitalocean/` scripts are kept solely as provider-agnostic Linux-GPU setup
reused by GCP.
63
64     ## Workflow conventions
65     - **Branch off `master` ; ONE PR per work item; NO PR stacking.** Don't open a PR unless
asked.
66     - Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.10+ (the `trackers`
package requires ?3.10); `ffmpeg` required.
67     - Never push to the default branch directly; descriptive commit messages.
68     - **Substantial design/project work gets a tracked project doc** following
`docs/PROJECT_DOC_TEMPLATE.md` (status header · decisions-locked · progress tracker ·
definition-of-done; archived on completion). Precedents: `docs/RALLY_QUALITY_RESEARCH.md` §7,
`docs/GOLDEN_REGRESSION_FIXTURES.md`.
69     - **Launch / "bit-flip" PRs** (a default-flip that drives a tracked project to terminal
state) MUST, in the SAME PR, run the `docs/DOC_STATUS.md` §2 archival due-diligence on that
project's docs (verify-vs-code ? flip status + completion note ? update inbound refs ? keep ADR
lineage ? `git mv` to `docs/archives/` ? record in DOC_STATUS §3), and fix any internal-link
breakage the move causes. Enforced by `.github/PULL_REQUEST_TEMPLATE.md` + the `link-check` CI
job (`python -m tools.check_md_links`).
70     - ?? Hand the heaviest machine-side checks (real video, the full GPU/WSL/WASB suite) to
the owner,
71     who has the golden corpus + labelled videos; verify pure-logic paths locally.
72

```

4. user (2026-07-02T04:45:59.303Z)

```

Exit code 128
fatal: Cannot fast-forward to multiple branches.

```

5. user (2026-07-02T04:46:05.588Z)

```

master
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)

```

6. user (2026-07-02T04:46:12.379Z)

```

e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
0
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer e410136 [master]

```

7. user (2026-07-02T04:46:21.251Z)

```

Preparing worktree (new branch 'fix/upload-dir-app-home')

```

branch 'fix/upload-dir-app-home' set up to track 'origin/master'.
HEAD is now at e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)

8. user (2026-07-02T04:46:30.120Z)

```
1      """Chunked upload (B2, PR-2) ? extracted from backend.main into its own router.
2
3      Browser uploads arrive in sequential parts ? security.max_part_mb (free-tier edge
4      proxies cap request bodies at ~100 MB ? HOSTING_PLAN $2), are assembled server-side
5      under security.upload_dir, then submitted to /api/process by the returned path.
6      Upload state is in-process by design (this is the single-box alpha): a server restart
7      aborts partial uploads and the client re-inits. Per-user partitioning of upload_dir
8      arrives with PR-3 identity scoping.
9
10     The startup config lives on ``backend.main.global_config`` (set ONLY in ``main()``).
11     ``_upload_cfg`` resolves it through the ``backend.main`` module object so it always
12     reads the live value the server set at startup (and so tests that monkeypatch
13     ``backend.main.global_config`` flow through unchanged).
14     """
15
16     import os
17     import threading
18     import uuid
19     from typing import Any, Dict, Optional
20
21     from fastapi import APIRouter, HTTPException, Request
22     from pydantic import BaseModel
23
24     from backend.utils.freespace import require_free_bytes # HTTP-507 disk guard (P2, D11)
25     from backend.utils.hashing import compute_video_id # canonical file-identity hashing
26     from backend.utils.paths import safe_output_filename
27
28     router = APIRouter()
29
30     _uploads: Dict[str, Dict[str, Any]] = {}
31     _uploads_lock = threading.Lock()
32
33
34     class UploadInitRequest(BaseModel):
35         filename: str
36         total_size_bytes: Optional[int] = None
37
38
39     class UploadCompleteRequest(BaseModel):
40         total_parts: int
41
42
43     def _upload_cfg():
44         # Resolved lazily (import inside the function) to read the LIVE startup config that
45         # main() sets on the module global, and to avoid a circular import at module load
46         # (main.py imports this router; this module must not import main at import time).
47         import backend.main as _main
48
49         sec = getattr(_main.global_config, "security", None)
50         upload_dir = getattr(sec, "upload_dir", None) or "output/uploads"
51         max_bytes = int(getattr(sec, "max_upload_mb", 4096) or 4096) * 1024 * 1024
52         part_bytes = int(getattr(sec, "max_part_mb", 95) or 95) * 1024 * 1024
53         exts = [
54             e.lower().lstrip(".")
55             for e in (getattr(sec, "allowed_upload_extensions", None) or ["mp4", "mov"])
56         ]
57         return upload_dir, max_bytes, part_bytes, exts
58
59
```

```

60 def _abort_upload(upload_id: str) -> None:
61     with _uploads_lock:
62         st = _uploads.pop(upload_id, None)
63     if st:
64         try:
65             os.remove(st["tmp_path"])
66         except OSError:
67             pass
68
69
70 @router.post("/api/upload/init")
71 def upload_init(req: UploadInitRequest):
72     """Start a chunked upload. Returns upload_id; PUT sequential parts next."""
73     upload_dir, max_bytes, part_bytes, exts = _upload_cfg()
74     try:
75         name = safe_output_filename(req.filename)
76     except ValueError as e:
77         raise HTTPException(status_code=400, detail=str(e)) from e
78     ext = os.path.splitext(name)[1].lower().lstrip(".")
79     if ext not in exts:
80         raise HTTPException(
81             status_code=400,
82             detail=f"Extension '{ext}' not allowed. Allowed: {sorted(exts)}",
83         )
84     if req.total_size_bytes is not None and req.total_size_bytes > max_bytes:
85         raise HTTPException(
86             status_code=413,
87             detail=f"File exceeds the {max_bytes // (1024 * 1024)} MB upload limit.",
88         )
89     os.makedirs(upload_dir, exist_ok=True)
90     # P2 (D11): fail fast with 507 if the box can't hold the declared upload. Best-
91     # effort ? an
92     # undeclared size (total_size_bytes=None) or an unreadable disk skips the guard
93     # (never blocks a
94     # legit upload); the per-part total cap in upload_part is the hard backstop either
95     # way.
96     require_free_bytes(upload_dir, req.total_size_bytes, stage="upload")
97     upload_id = uuid.uuid4().hex
98     tmp_path = os.path.join(upload_dir, f".partial-{upload_id}")
99     open(tmp_path, "wb").close()
100     with _uploads_lock:
101         _uploads[upload_id] = {
102             "filename": name,
103             "tmp_path": tmp_path,
104             "parts": 0,
105             "bytes": 0,
106         }
107     return {
108         "upload_id": upload_id,
109         "max_part_mb": part_bytes // (1024 * 1024),
110         "next_part": f"/api/upload/{upload_id}/part/0",
111     }
112
113 @router.put("/api/upload/{upload_id}/part/{index}")
114 async def upload_part(upload_id: str, index: int, request: Request):
115     """Append one part (raw bytes body). Parts are strictly sequential (0,1,2,...)."""
116     _, max_bytes, part_bytes, _ = _upload_cfg()
117     with _uploads_lock:
118         st = _uploads.get(upload_id)
119     if st is None:
120         raise HTTPException(status_code=404, detail=f"Unknown upload_id '{upload_id}'")
121     if index != st["parts"]:
122         raise HTTPException(
123             status_code=409,

```

```

122         detail=f"Out-of-order part: expected {st['parts']], got {index} (parts are
sequential)",
123     )
124
125     received = 0
126     overflow = None
127     with open(st["tmp_path"], "ab") as fh:
128         async for chunk in request.stream():
129             received += len(chunk)
130             if received > part_bytes:
131                 overflow = (
132                     f"Part exceeds the {part_bytes // (1024 * 1024)} MB per-part limit."
133                 )
134                 break
135             if st["bytes"] + received > max_bytes:
136                 overflow = (
137                     f"Upload exceeds the {max_bytes // (1024 * 1024)} MB total limit."
138                 )
139                 break
140             fh.write(chunk)
141     if overflow:
142         _abort_upload(upload_id)
143         raise HTTPException(
144             status_code=413, detail=overflow + " Upload aborted ? re-init to retry."
145         )
146
147     with _uploads_lock:
148         st["parts"] += 1
149         st["bytes"] += received
150     return {
151         "upload_id": upload_id,
152         "received_parts": st["parts"],
153         "received_bytes": st["bytes"],
154     }
155
156
157     @router.post("/api/upload/{upload_id}/complete")
158     def upload_complete(upload_id: str, req: UploadCompleteRequest):
159         """Finalize: verify part count, move into place, return {path, video_id} for
/api/process."""
160         upload_dir, _, _, _ = _upload_cfg()
161         with _uploads_lock:
162             st = _uploads.get(upload_id)
163         if st is None:
164             raise HTTPException(status_code=404, detail=f"Unknown upload_id '{upload_id}'")
165         if st["parts"] == 0 or st["parts"] != req.total_parts:
166             raise HTTPException(
167                 status_code=409,
168                 detail=f"Expected {req.total_parts} part(s), received {st['parts']}.",
169             )
170         final_path = os.path.join(upload_dir, st["filename"])
171         if os.path.exists(
172             final_path
173         ): # never overwrite an earlier upload with the same name
174             stem, ext = os.path.splitext(st["filename"])
175             final_path = os.path.join(upload_dir, f"{stem}-{upload_id[:8]}{ext}")
176         os.replace(st["tmp_path"], final_path)
177         with _uploads_lock:
178             _uploads.pop(upload_id, None)
179         video_id = compute_video_id(final_path)
180         return {
181             "path": os.path.abspath(final_path),
182             "video_id": video_id,
183             "size_bytes": os.path.getsize(final_path),
184             "next": "POST /api/process with this path",

```



```

185         }
186
187
188     @router.delete("/api/upload/{upload_id}")
189     def upload_abort(upload_id: str):
190         """Abort an in-flight upload and delete its partial file."""
191         with _uploads_lock:
192             known = upload_id in _uploads
193             if not known:
194                 raise HTTPException(status_code=404, detail=f"Unknown upload_id '{upload_id}'")
195             _abort_upload(upload_id)
196             return {"status": "aborted", "upload_id": upload_id}
197

```

9. user (2026-07-02T04:46:31.063Z)

```

1     """Resolved application-data HOME for a packaged, batteries-included install.
2
3     PACKAGING_PLAN.md P0a. Today every piece of mutable state ? ``config.json``, ``.env``,
4     the SQLite DB, ``output/`` ? is resolved RELATIVE TO THE CURRENT WORKING DIRECTORY, and
5     ``scripts/doctor.py`` writes ``config.json`` to the repo root while the server reads it
6     from
7     the CWD. Launched from an OS app menu (CWD ? repo root) those disagree and the user
8     silently
9     "loses" their index. This module introduces ONE resolver every state path roots on.
10
11     Design contract ? **DEFAULT IS BYTE-IDENTICAL TO TODAY**:
12     * ``RALLY_APP_HOME`` env set ? that directory is the app home (the packaged launcher
13     sets it).
14     * ``RALLY_APP_HOME`` unset ? ``Path(".")`` i.e. the CWD, exactly the pre-P0a
15     behaviour.
16
17     The OS-default location (``%APPDATA%`` / ``~/Library/Application Support`` /
18     ``~/local/share``)
19     is computed by :func:`default_os_app_home` but is **NOT** auto-selected here ? the
20     launcher (P2)
21     calls it and exports ``RALLY_APP_HOME`` explicitly. Keeping the resolver opt-in means
22     importing
23     this module has zero behavioural effect on the repo/dev/CI/test paths (no surprise
24     writes into a
25     real user profile during a test run).
26
27     stdlib-only (os, platform, pathlib) so it is safe to import at ``backend.main`` line 1 ?
28     before
29     the heavier pydantic-backed ``backend.config`` package ? keeping the early
30     ``load_dotenv`` path light.
31     """
32
33     from __future__ import annotations
34
35     import os
36     import platform
37     from pathlib import Path
38
39     #: Environment variable the packaged launcher sets to pin the app-data home.
40     APP_HOME_ENV = "RALLY_APP_HOME"
41
42     #: OS-app-dir leaf used by :func:`default_os_app_home` (the product brand).
43     APP_DIR_NAME = "Khelsutra"
44
45     def app_home() -> Path:
46         """The resolved application-data home.
47
48         ``RALLY_APP_HOME`` (``~`` expanded) when set; otherwise ``Path(".")`` ? the CWD,

```

```

byte-identical
40         to the pre-P0a behaviour. Does NOT create the directory (callers create on write).
41         ""
42         env = os.environ.get(APP_HOME_ENV)
43         if env and env.strip():
44             return Path(env.strip()).expanduser()
45         return Path(".")
46
47
48     def app_path(*parts: str | os.PathLike[str]) -> Path:
49         """A path under :func:`app_home` (e.g. ``app_path("output")``)."""
50         return app_home().joinpath(*[str(p) for p in parts])
51
52
53     def default_config_path() -> Path:
54         """`<app-home>/config.json` ? the config file the loader reads and ``--setup``
writes.
55
56         With ``RALLY_APP_HOME`` unset this is ``Path("config.json")`` (CWD-relative),
unchanged.
57         ""
58         return app_home() / "config.json"
59
60
61     def default_env_path() -> Path:
62         """`<app-home>/env` ? the dotenv file the server prefers (e.g.
`GEMINI_API_KEY`)."""
63         return app_home() / ".env"
64
65
66     def env_file_to_load() -> Path | None:
67         """The ``.env`` the server should load, or ``None`` meaning "use python-dotenv's
DEFAULT
68         (frame-relative walk-up) search" ? exactly the pre-P0a bare ``load_dotenv()``
behaviour.
69
70         Returns the app-home ``.env`` ONLY when ``RALLY_APP_HOME`` is set AND that file
exists. With the
71         env UNSET this returns ``None`` so the caller keeps the historical search and does
NOT newly
72         prefer a stray CWD ``.env`` (bare ``load_dotenv()`` walks up from the caller file,
not the CWD).
73         ""
74         home = app_home()
75         if (
76             str(home) == "."
77         ): # env unset (or pinned to CWD) ? historical load_dotenv() search
78             return None
79         candidate = home / ".env"
80         return candidate if candidate.exists() else None
81
82
83     def resolve_output_dir(output_dir: str) -> str:
84         """Root a RELATIVE ``output_dir`` under :func:`app_home`; pass an ABSOLUTE one
through.
85
86         With ``RALLY_APP_HOME`` UNSET the input is returned VERBATIM (the default
``"output"`` when
87         empty) ? byte-identical to the pre-P0a code that stored ``--output-dir`` as-is (no
pathlib
88         normalisation of ``./x`` / separators / trailing slash). With it SET, a relative dir
like
89         ``"output"`` becomes ``<app-home>/output`` so the SQLite DB and reels live with the
rest of the
90         app's state; an absolute path ? or a Windows drive-anchored one (``C:\\x`` /

```

```

``C:x``) ? passes
91     through untouched.
92     """
93     home = app_home()
94     if str(home) == ".": # env unset (or pinned to CWD) ? verbatim, today's behaviour
95         return output_dir or "output"
96     if not output_dir:
97         return str(home / "output")
98     if os.path.isabs(output_dir) or os.path.splitdrive(output_dir)[0]:
99         return output_dir
100    return str(home / output_dir)
101
102
103    def default_os_app_home(app_name: str = APP_DIR_NAME) -> Path:
104        """The OS-conventional per-user app-data dir (for the packaged LAUNCHER to export as
105        ``RALLY_APP_HOME`` ? NOT auto-selected by :func:`app_home`).
106
107        * Windows ? ``%APPDATA%\<app_name>`` (``~/AppData/Roaming/<app_name>`` if unset)
108        * macOS   ? ``~/Library/Application Support/<app_name>``
109        * Linux/other ? ``$XDG_DATA_HOME/<app_name>`` (``~/.local/share/<app_name>`` if
unset)
110        """
111        system = platform.system()
112        if system == "Windows":
113            base = os.environ.get("APPDATA") or str(Path.home() / "AppData" / "Roaming")
114        elif system == "Darwin":
115            base = str(Path.home() / "Library" / "Application Support")
116        else:
117            base = os.environ.get("XDG_DATA_HOME") or str(Path.home() / ".local" / "share")
118        return Path(base) / app_name
119

```

10. user (2026-07-02T04:46:43.682Z)

File does not exist. Note: your current working directory is C:\Users\avidu\Projects\khelesutra-guru\badminton-highlight-indexer.

11. user (2026-07-02T04:46:45.023Z)

```

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\backend\api\uploads.py:5:under security.upload_dir, then submitted to /api/process by the
returned path.
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\backend\api\uploads.py:7:aborts partial uploads and the client re-inits. Per-user
partitioning of upload_dir
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-dir\backend\api\uploads.py:50:
upload_dir = getattr(sec, "upload_dir", None) or "output/uploads"
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-dir\backend\api\uploads.py:57:
return upload_dir, max_bytes, part_bytes, exts
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-dir\backend\api\uploads.py:73:
upload_dir, max_bytes, part_bytes, exts = _upload_cfg()
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-dir\backend\api\uploads.py:89:
os.makedirs(upload_dir, exist_ok=True)
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-dir\backend\api\uploads.py:93:
require_free_bytes(upload_dir, req.total_size_bytes, stage="upload")
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-dir\backend\api\uploads.py:95:
tmp_path = os.path.join(upload_dir, f".partial-{upload_id}")

```

```

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\backend\api\uploads.py:160:      upload_dir, _, _ = _upload_cfg()
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\backend\api\uploads.py:170:      final_path = os.path.join(upload_dir, st["filename"])
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\backend\api\uploads.py:175:      final_path = os.path.join(upload_dir,
f"{stem}-{upload_id[:8]}{ext}")
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\backend\config\models.py:335:      # contain upload_dir or /api/process will reject the
uploaded paths.
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\backend\config\models.py:336:      upload_dir: str = "output/uploads"
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\backend\utils\freespace.py:4: ``upload_dir``, a fetch-to-scratch of a bucket/Drive
source (a full local copy under the temp
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\backend\utils\freespace.py:32:      ancestor (an ``upload_dir``/scratch target may not exist
yet at guard time)."""
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tools\cleanup_caches.py:5:( ``security.upload_dir``, default ``output/uploads``) ? classifies
each entry as
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tools\cleanup_caches.py:30:automatically when ``wsl`` is unavailable). ``--uploads-dir``
must match ``security.upload_dir``
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tools\cleanup_caches.py:56:      # The upload landing zone (security.upload_dir). Uploaded
source videos and the
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tools\cleanup_caches.py:274:      Flat by design today (uploads.py writes
``<upload_dir>/<filename>`` plus
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tools\cleanup_caches.py:403:      help="upload landing zone to sweep (default <output-
dir>/uploads; match security.upload_dir)",
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\docs\CODE_MAP.md:115:[Omitted long matching line]
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\docs\COMPUTE_DECOUPLED_SERVING\README.md:170:[Omitted long matching line]
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\docs\VIDEO_LOCALITY_MODEL.md:13:assembles the full file under ``security.upload_dir`` (default
``output/uploads``) on
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tests\test_upload_download.py:31:      cfg.security.upload_dir = str(tmp_path / "up")
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tests\test_upload_download.py:125:      f for f in os.listdir(cfg.security.upload_dir) if
f.startswith(".partial-")
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tests\test_upload_download.py:206:      f for f in os.listdir(cfg.security.upload_dir) if

```

```
f.startswith(".partial-")
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tests\test_upload_download.py:232:      f for f in os.listdir(cfg.security.upload_dir) if
f.startswith(".partial-")
```

12. user (2026-07-02T04:46:52.382Z)

```
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-dir\scripts\install.py
```

13. user (2026-07-02T04:46:53.821Z)

```
320      # Third-party HTTP/RPC client loggers (httpx, httpcore, urllib3, google-genai,
321      # google.auth, grpc) emit one INFO line per request ? dozens per Gemini run, which
322      # buries our own [rally]/[compile] lines in run.log during manual QA. Default raises
323      # them to WARNING; set true to restore full chatter for request-flow debugging.
324      verbose_http: bool = False
325
326
327      class SecurityConfig(BaseModel):
328          # When set, /api/process video_path must resolve under this directory (hosted/tenant
329          # mode). Default None preserves the single-user local 'any local file' workflow.
330          allowed_video_root: Optional[str] = None
331
332          # --- Chunked upload (B2, PR-2) ---
333          # Landing zone for browser uploads (assembled from parts). Per-user subdirectories
334          # arrive with PR-3 identity scoping. ? In hosted mode, set allowed_video_root to
335          # contain upload_dir or /api/process will reject the uploaded paths.
336          upload_dir: str = "output/uploads"
337          # Whole-file cap (default 4 GB) and per-part cap. Parts stay under ~100 MB because
338          # free-tier edge proxies cap request bodies there (HOSTING_PLAN §2).
339          max_upload_mb: int = 4096
340          max_part_mb: int = 95
341          allowed_upload_extensions: List[str] = ["mp4", "mov"]
342
343          # --- M9 edge auth (Cloudflare Access, D9) ---
344          # DEFAULT-OFF: edge_auth_enabled=False ? no JWT is required/verified (the local
single-user
345          # path is unchanged). When True, /api/process requires + verifies the `Cf-Access-
Jwt-Assertion`
346          # header against the team JWKS (so a direct hit to the Cloud Run URL can't spoof the
identity
347          # header) and tags the job with the verified user_id (owner_id). The team domain
(e.g.
348          # ``https://<team>.cloudflareaccess.com``) supplies the JWKS + issuer; the AUD is
the CF Access
349          # application's audience tag. NO secrets here ? these are public identifiers.
350          edge_auth_enabled: bool = False
351          cf_team_domain: str = (
352              "" # e.g. https://<team>.cloudflareaccess.com (issuer + /certs JWKS)
353          )
354          cf_access_aud: str = "" # the CF Access application AUD tag the token must carry
355          # (CORS allow-origins is set via the RALLY_CORS_ALLOW_ORIGINS env var ? read at
import in
356          # backend/main.py so importing the app does no filesystem I/O; default [""].
Credentials stay
357          # OFF ? Cf-Access is a header, not a cookie.)
358
359
```

14. user (2026-07-02T04:47:00.435Z)

```
225      dst_uninstall = app_home / "uninstall.py"
```

```

226         if src_uninstall.exists():
227             shutil.copyfile(src_uninstall, dst_uninstall)
228             created.append(dst_uninstall)
229
230         print("\n[3/3] writing the uninstall manifest?")
231         manifest_path = app_home / MANIFEST_NAME
232         created.append(manifest_path)
233         manifest = build_manifest(
234             method=method,
235             app_home=app_home,
236             shortcut=shortcut,
237             created=created,
238             timestamp=datetime.now(timezone.utc).isoformat(timespec="seconds"),
239         )
240         manifest_path.write_text(json.dumps(manifest, indent=2) + "\n", encoding="utf-8")
241
242         print("\n? Installed. To launch KhelSutra:")
243         print(f"    run the launcher {shortcut}    (or `indexer --app` from a shell)")
244         print(
245             "    The first run seeds a truly-local 'motion' config (offline, no key/GPU
needed)."
246         )
247         if indexer_exe is None:
248             print(
249                 "    ? `indexer` isn't on this shell's PATH yet ? the launcher uses the bare
name. If it"
250             )
251             print(
252                 "    fails, reopen your terminal (uv just ran `update-shell`), add the
install bin dir to"
253             )
254             print(
255                 "    PATH, or launch with `python -m backend.main --app` from the install
environment."
256             )
257             print(
258                 f"    To uninstall:  python \"{dst_uninstall}\"    (add --purge to also delete your
data)"
259             )
260             return 0
261
262
263     if __name__ == "__main__":
264         raise SystemExit(main())
265

```

15. user (2026-07-02T04:47:01.839Z)

```

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tests\test_app_home.py:38:def test_resolve_output_dir_unset_is_verbatim_for_relative():
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-dir\tests\test_app_home.py:42:
assert ah.resolve_output_dir(v) == v
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-dir\tests\test_app_home.py:43:
assert ah.resolve_output_dir("") == "output" # empty ? the historical default
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tests\test_app_home.py:79:def test_resolve_output_dir_set(monkeypatch, tmp_path):
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-dir\tests\test_app_home.py:81:
assert ah.resolve_output_dir("output") == str(tmp_path / "output")
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-

```

```

dir\tests\test_app_home.py:84: def test_resolve_output_dir_absolute_passthrough(monkeypatch,
tmp_path):
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-dir\tests\test_app_home.py:91:
assert ah.resolve_output_dir(abs_dir) == abs_dir
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tests\test_app_home.py:94: def
test_resolve_output_dir_empty_falls_back_to_home_output(monkeypatch, tmp_path):
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-dir\tests\test_app_home.py:96:
assert ah.resolve_output_dir("") == str(tmp_path / "output")
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tests\test_app_home.py:177: # ---- resolve_output_dir passthrough for absolute / Windows
drive-anchored inputs ----
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tests\test_app_home.py:180: def test_resolve_output_dir_set_absolute_passthrough(monkeypatch,
tmp_path):
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tests\test_app_home.py:183: assert ah.resolve_output_dir(abs_p) == abs_p
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tests\test_app_home.py:186: def
test_resolve_output_dir_set_windows_drive_passthrough(monkeypatch, tmp_path):
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tests\test_app_home.py:190: assert ah.resolve_output_dir("C:out") == "C:out"
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tests\test_app_home.py:191: assert ah.resolve_output_dir("C:\\out") == "C:\\out"
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\docs\PACKAGING_PLAN.md:100: | **P0a** | **App-data HOME** (`RALLY_APP_HOME` env + per-OS
default via `default_os_app_home`) threaded through `default_config_path()`,
`load_dotenv(env_file_to_load())`, `resolve_output_dir` (output/DB), package-relative static
mount; **unified `doctor.py` write with the server read**. Byte-identical when unset; 4-lens
adversarial review folded. | ? | No | ? | [#293](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/293) |
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\backend\utils\app_home.py:83: def resolve_output_dir(output_dir: str) -> str:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-dir\backend\main.py:25: from
backend.utils.app_home import env_file_to_load, resolve_output_dir
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-dir\backend\main.py:192:
out_dir = resolve_output_dir(
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-dir\backend\main.py:199: #
unset, resolve_output_dir("output") == "output" (CWD-relative) ? byte-identical to before.
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-dir\backend\main.py:199:
global_config.output.output_dir = resolve_output_dir(args.output_dir)
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tests\test_server_entrypoint.py:119: SQLite DB must land under the app HOME ? not the
launch CWD ? proving resolve_output_dir is
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-upload-
dir\tests\test_server_entrypoint.py:165: "DB was NOT created under RALLY_APP_HOME ?
resolve_output_dir is not wired into main() init.\n"

```

16. user (2026-07-02T04:47:09.221Z)

```
1      """Chunked upload (B2) + highlight download (B3) ? PR-2 of the UI alpha plan.
2
3      Contract under test:
4      - POST /api/upload/init ? {upload_id}; extension allowlist + size caps enforced
5      - PUT /api/upload/{id}/part/{n} ? strictly sequential parts; per-part/total caps abort
(413)
6      - POST /api/upload/{id}/complete ? assembled file, MD5 == compute_video_id, no overwrite
7      - DELETE /api/upload/{id} ? abort removes partial state
8      - GET /api/highlights/{video_id}/{filename} ? streams from output_dir; traversal-proof
9      """
10
11     import os
12     from unittest.mock import patch
13
14     import pytest
15
16     pytest.importorskip("httpx") # TestClient needs httpx; skip cleanly where absent
17     from fastapi.testclient import TestClient
18
19     import backend.main as m
20     from backend.api import uploads as uploads_api
21     from backend.config.models import AppConfig
22     from backend.infrastructure.database import Database
23     from backend.utils.hashing import compute_video_id
24
25
26     @pytest.fixture
27     def env(tmp_path, monkeypatch):
28         db = Database(str(tmp_path / "t.db"))
29         cfg = AppConfig()
30         cfg.output.output_dir = str(tmp_path / "out")
31         cfg.security.upload_dir = str(tmp_path / "up")
32         os.makedirs(cfg.output.output_dir, exist_ok=True)
33         monkeypatch.setattr(
34             uploads_api, "_uploads", {}
35         ) # isolate upload state per test (registry moved to backend.api.uploads)
36         app = m.create_app(cfg, db)
37         client = TestClient(app)
38         return client, db, tmp_path, cfg
39
40
41     def _init(client, filename="match.mp4", total=None):
42         body = {"filename": filename}
43         if total is not None:
44             body["total_size_bytes"] = total
45         return client.post("/api/upload/init", json=body)
46
47
48     # --- init -----
49
50
51     def test_init_rejects_bad_extension(env):
52         client = env[0]
53         r = _init(client, "malware.exe")
54         assert r.status_code == 400
55         assert "not allowed" in r.json()["detail"]
56
57
58     def test_init_rejects_traversal_filename(env):
59         client = env[0]
60         assert _init(client, "../evil.mp4").status_code == 400
61         assert _init(client, "a/b.mp4").status_code == 400
62
```



```

63
64 def test_init_rejects_oversize_declared_total(env):
65     client, db, tmp_path, cfg = env
66     cfg.security.max_upload_mb = 1
67     r = _init(client, total=2 * 1024 * 1024)
68     assert r.status_code == 413
69
70
71 def test_init_507_when_disk_too_full_for_declared_total(env):
72     # P2 (D11): a declared upload that won't fit fails fast with 507 (not a mid-upload
73     crash).
74     import types
75
76     from backend.utils import freespace
77
78     client = env[0]
79     with patch.object(
80         freespace.shutil,
81         "disk_usage",
82         return_value=types.SimpleNamespace(total=0, used=0, free=1 * 1024 * 1024),
83     ):
84         r = _init(client, total=50 * 1024 * 1024)
85         assert r.status_code == 507
86         assert "space" in r.json()["detail"].lower()
87
88 def test_init_ok_when_disk_has_room(env):
89     import types
90
91     from backend.utils import freespace
92
93     client = env[0]
94     with patch.object(
95         freespace.shutil,
96         "disk_usage",
97         return_value=types.SimpleNamespace(total=0, used=0, free=500 * 1024 * 1024),
98     ):
99         r = _init(client, total=50 * 1024 * 1024)
100         assert r.status_code == 200
101
102
103 # --- full lifecycle -----
104
105
106 def test_roundtrip_assembles_parts_and_matches_md5(env):
107     client, db, tmp_path, cfg = env
108     up = _init(client).json()
109     parts = [b"a" * 1000, b"b" * 2000, b"c" * 500]
110     for i, blob in enumerate(parts):
111         r = client.put(f"/api/upload/{up['upload_id']}/part/{i}", content=blob)
112         assert r.status_code == 200
113         assert r.json()["received_parts"] == i + 1
114     done = client.post(
115         f"/api/upload/{up['upload_id']}/complete", json={"total_parts": 3}
116     )
117     assert done.status_code == 200
118     body = done.json()
119     assert body["size_bytes"] == 3500
120     with open(body["path"], "rb") as fh:
121         assert fh.read() == b"".join(parts)
122     assert body["video_id"] == compute_video_id(body["path"])
123     # partial file cleaned up, registry entry gone
124     assert not [
125         f for f in os.listdir(cfg.security.upload_dir) if f.startswith(".partial-")
126 ]

```

```

127     assert (
128         client.put(f"/api/upload/{up['upload_id']}/part/3", content=b"x").status_code
129         == 404
130     )
131
132
133 def test_same_filename_twice_does_not_overwrite(env):
134     client = env[0]
135     paths = []
136     for content in (b"first", b"second"):
137         up = _init(client).json()
138         client.put(f"/api/upload/{up['upload_id']}/part/0", content=content)
139         done = client.post(
140             f"/api/upload/{up['upload_id']}/complete", json={"total_parts": 1}
141         )
142         paths.append(done.json()["path"])
143     assert paths[0] != paths[1]
144     with open(paths[0], "rb") as fh:
145         assert fh.read() == b"first" # first upload untouched
146     with open(paths[1], "rb") as fh:
147         assert fh.read() == b"second"
148
149
150 # --- sequencing & caps -----
151
152
153 def test_out_of_order_part_is_409(env):
154     client = env[0]
155     up = _init(client).json()
156     assert (
157         client.put(f"/api/upload/{up['upload_id']}/part/1", content=b"x").status_code
158         == 409
159     )
160
161
162 def test_unknown_upload_id_is_404(env):
163     client = env[0]
164     assert client.put("/api/upload/nope/part/0", content=b"x").status_code == 404
165     assert (
166         client.post("/api/upload/nope/complete", json={"total_parts": 1}).status_code
167         == 404
168     )
169     assert client.delete("/api/upload/nope").status_code == 404
170
171
172 def test_complete_with_wrong_part_count_is_409(env):
173     client = env[0]
174     up = _init(client).json()
175     client.put(f"/api/upload/{up['upload_id']}/part/0", content=b"x")
176     assert (
177         client.post(
178             f"/api/upload/{up['upload_id']}/complete", json={"total_parts": 2}
179         ).status_code
180         == 409
181     )
182     # zero-part complete is also a 409
183     up2 = _init(client).json()
184     assert (
185         client.post(
186             f"/api/upload/{up2['upload_id']}/complete", json={"total_parts": 0}
187         ).status_code
188         == 409
189     )
190
191

```

```

192 def test_oversize_part_aborts_upload(env):
193     client, db, tmp_path, cfg = env
194     cfg.security.max_part_mb = 1
195     up = _init(client).json()
196     r = client.put(
197         f"/api/upload/{up['upload_id']}/part/0", content=b"x" * (1024 * 1024 + 10)
198     )
199     assert r.status_code == 413
200     # aborted: id unknown afterwards, no partials on disk
201     assert (
202         client.put(f"/api/upload/{up['upload_id']}/part/0", content=b"x").status_code
203         == 404
204     )
205     assert not [
206         f for f in os.listdir(cfg.security.upload_dir) if f.startswith(".partial-")
207     ]
208
209
210 def test_total_cap_across_parts_aborts_upload(env):
211     client, db, tmp_path, cfg = env
212     cfg.security.max_upload_mb = 1
213     up = _init(client).json()
214     ok = client.put(
215         f"/api/upload/{up['upload_id']}/part/0", content=b"x" * (700 * 1024)
216     )
217     assert ok.status_code == 200
218     r = client.put(f"/api/upload/{up['upload_id']}/part/1", content=b"x" * (700 * 1024))
219     assert r.status_code == 413
220
221
222 def test_abort_endpoint_removes_state(env):
223     client, db, tmp_path, cfg = env
224     up = _init(client).json()
225     client.put(f"/api/upload/{up['upload_id']}/part/0", content=b"x")
226     assert client.delete(f"/api/upload/{up['upload_id']}").status_code == 200
227     assert (
228         client.put(f"/api/upload/{up['upload_id']}/part/1", content=b"x").status_code
229         == 404
230     )
231     assert not [
232         f for f in os.listdir(cfg.security.upload_dir) if f.startswith(".partial-")
233     ]
234
235
236 # --- download (B3)
-----
237
238
239 def test_download_unknown_video_404(env):
240     client = env[0]
241     assert client.get("/api/highlights/nope/reel.mp4").status_code == 404
242
243
244 def test_download_traversal_filename_rejected(env):
245     client, db, tmp_path, cfg = env
246     db.add_video("vid", "/x.mp4", "processed", [])
247     r = client.get("/api/highlights/vid/..%2Fevil.mp4")
248     assert r.status_code in (
249         400,
250         404,
251     ) # 400 from safe-name; 404 if the router normalizes
252     r2 = client.get("/api/highlights/vid/a%5Cb.mp4") # backslash separator
253     assert r2.status_code == 400
254
255

```

```

256     def test_download_missing_file_404(env):
257         client, db, tmp_path, cfg = env
258         db.add_video("vid", "/x.mp4", "processed", [])
259         assert client.get("/api/highlights/vid/reel.mp4").status_code == 404
260
261
262     def test_download_streams_compiled_file(env):
263         client, db, tmp_path, cfg = env
264         db.add_video("vid", "/x.mp4", "processed", [])
265         blob = b"\x00\x00fake-mp4-bytes" * 100
266         with open(os.path.join(cfg.output.output_dir, "reel.mp4"), "wb") as fh:
267             fh.write(blob)
268         r = client.get("/api/highlights/vid/reel.mp4")
269         assert r.status_code == 200
270         assert r.content == blob
271         assert r.headers["content-type"].startswith("video/mp4")
272         assert "reel.mp4" in r.headers.get("content-disposition", "")
273

```

17. user (2026-07-02T04:47:13.403Z)

20:## 0. REPO LAYOUT ? where new files go (READ BEFORE committing a new file)

18. user (2026-07-02T04:47:20.233Z)

```

20     ## 0. REPO LAYOUT ? where new files go (READ BEFORE committing a new file)
21
22     Top-level tree and the placement rule for each kind of file. When unsure, prefer an
existing
23     registry / extension point (``) over a new top-level file.
24
25     | You're adding? | Put it in? | Notes |
26     |---|---|---|
27     | Backend / runtime code | `backend/<subsystem>/` |
`pipeline/{segmenters,detectors,validators}`, `providers/`, `sports/`, `annotations/`,
`infrastructure/`, `reporting/`, `stitcher/`, `utils/`, `config/`, `features/`, `api/`. Most
additions are register a class in the relevant `` registry, not a new module tree. |
28     | Eval LIBRARY (importable, pure ? metrics, calibration, a scorer, a feature) |
`backend/eval/` | Pure core, no side-effects at import; CLI/I-O lives in a separate driver.
These are imported widely (e.g. `calibrate_wasb` by 8 sites) ? they are library, not
scripts; do NOT move them out. |
29     | Standalone run-driver / one-off entry CLI (has `__main__`, NOT imported by app
code) | scripts/ | e.g. `scripts/run_phase3_eval.py`, `scripts/run_process_and_stitch.py`.
Run as `python scripts/<name>.py`. If it needs `load_dotenv()/sys.path` before importing
backend, add `scripts/<name>.py" = ["E402"]` to `ruff.toml`. |
30     | Ops / maintenance tool (not part of the served app ? cache GC, batch admin) |
`tools/` | Run as `python -m tools.<name>` (it's an importable package). e.g.
`tools/cleanup_caches.py`. Keep the destructive decision path pure + unit-tested. |
31     | Training code (model fitting) | `training/` | Behind the CI-enforced import wall
(ADR-008) ? `backend.main` must not import it. Own `requirements-train.txt`. |
32     | A test | `tests/test_*.py` | pytest auto-discovers; the full-suite lane has a 70%
coverage floor. Pure/offline + mocked (no GPU/WSL/network). |
33     | A doc (design, plan, living log) | `docs/` | Index it in `docs/README.md`. Only
terminal-state (DONE/REJECTED) work moves to `docs/archives/`. |
34     | A throwaway repro / debug script | `scratch/` | Gitignored + excluded from
ruff/pytest. Never imported by app code. |
35     | A model artifact (weights + manifest) | `artifacts/<name>/<ver>/` | ADR-008:
manifest committed (provenance), weights gitignored. |
36     | Eval baseline / leaderboard JSON | `eval_baselines/` | Tracked (survives a clean
checkout); the no-regression gate reads it. |
37     | Committed regression fixture (video-free golden) |
`eval_baselines/fixtures/<slug>/` | Tracked, LF-pinned (`gitattributes`). Synthetic Tier-0
(or owner-cleared, de-attributed real). Drives the `golden_fixtures` characterization gate with
no video/GPU/DB. See `docs/GOLDEN_REGRESSION_FIXTURES.md`; mint via `python -m
backend.eval.golden_fixtures --freeze-synthetic` / `--freeze-real` (refusal-gated). |

```

```

38 | Pipeline run outputs (reels, trajectories, DBs, frames) | `output/` | **Gitignored.**
Per-video isolation is the tracked **`PER_VIDEO_WORKSPACE.md`** project (P0 helper shipped #264;
supersedes the archived `PER_VIDEO_OUTPUT_LAYOUT`). |
39
40 **Canonical files that stay at the repo ROOT** (referenced by CI / packaging / docs ? do
not move):
41 `pyproject.toml` (PEP 621 packaging, hatchling; `indexer` entry point) .
`run_all_tests.py` (CI: `ci.yml`) .
42 `run_regression.py` (the no-regression gate CLI, cited in `EXPERIMENT_HARNESS.md`) .
`run_eval.py`
43 (grandfathered ? imported by `tests/test_eval_harness.py`). The old diagnostic
`setup.py` moved to
44 `scripts/doctor.py` (it shadowed the build system); invoke via `python
scripts/doctor.py` or `indexer --setup`.
45 New standalone run-drivers go in **`scripts/`**, not the root.
46
47 ---
48
49 ## 1. PIPELINE
50
51 ### 1A. Runtime: video file -> highlight reel
52 ```
53 typed config load (built-in defaults -> config.json overrides; absent/parse-fail -> all-
defaults, never fatal)
54                                     @backend/config/loader.py::load_config ->
@backend/config/models.py::AppConfig
55     -> video_id = MD5(whole file)                                     @backend/utils/hashing.py::compute_video_id
(single shared helper, 64KB chunks; imported by main + all run_*.py ? no more per-entrypoint
copies)
56     -> validate
@backend/pipeline/validators/base.py::registry.run_all_with_context(video_path, global_config)
57     FormatValidator -> ctx['ffprobe_meta'] -> ResolutionFPSValidator -> PTSContinuity
-> SceneCut -> Blur -> Relevance (? ACTUAL registration/exec order ? see
@backend/pipeline/validators/__init__.py)
58     [validation FAIL -> persist status="failed", return early; report STILL emitted
in finally]
59     -> db.add_video(video_id, filepath, status, [r.dict()])
@backend/infrastructure/database.py
60     -> seg_name = req.segmenter or global_config.indexing.default_segmenter (fallback
'motion')
61     -> seg = segmenter_registry.get(seg_name)(db, global_config) ?
@backend/pipeline/segmenters/base.py::segmenter_registry (400 + available list if missing)
62     -> segments = seg.process_video(video_id, video_path, player_pool?, max_frames?)
63     [STOP POINT: segments-only ? predictions now in DB; eval/regression operate here
without stitching]
64     -> finally: emit_indexer_report(...) @backend/reporting/__init__.py (? ALWAYS runs
? even on validation failure / exception; NEVER raises; grafts response["reports"])
65     -> select segment ids -> [(start,end)] (+ stitching padding)
66     -> VideoCompiler.compile_highlights(filepath, time_pairs, out_name)
@backend/pipeline/stitcher/compiler.py
67     per-clip ffmpeg re-encode (libx264/superfast/crf22) -> concat -c copy ->
output/<name>.mp4
68     ```
69 Hybrid segmenter internal 4-phase (yolo_hybrid / tracknet_hybrid / wasb_hybrid ? same
shape):
70 ```
71 P1 chunk video (yolo only; tracknet/wasb = whole-video single pass, chunk_index=0)
72 -> P2 candidate "action windows" from motion/trajectory [STOP: candidates persisted
via db.add_candidate_segment]
73 -> P3 pad clip (ai_handoff_padding) -> ThreadPoolExecutor ->
provider.analyze_video(clip, prompt) ? provider_registry
74 -> P4 db.add_play_segment + db.link_candidate_to_segment
75 ```
76 Pure-AI path `gemini`: no P2; chunk-or-single -> provider -> heavy validate
(clamp/dedup) -> add_play_segment.

```

```
77     Legacy `motion`: pixel-diff -> segments + **fabricated** metadata/events (? not ground
truth).
78
79     ### 1B. WASB shuttle substrate (P2 of wasb_hybrid)
```

19. user (2026-07-02T04:47:20.907Z)

```
1     """Tests for the resolved application-data HOME (PACKAGING_PLAN.md P0a).
2
3     The load-bearing guarantee: with ``RALLY_APP_HOME`` UNSET, every resolver is byte-
identical to the
4     pre-P0a CWD-relative behaviour; with it SET, config/.env/output/DB all root under that
one home.
5     """
6
7     import os
8     from pathlib import Path
9
10    import pytest
11
12    from backend.config import loader
13    from backend.utils import app_home as ah
14
15
16    @pytest.fixture(autouse=True)
17    def _clear_app_home_env(monkeypatch):
18        """Each test starts with RALLY_APP_HOME unset unless it sets it explicitly."""
19        monkeypatch.delenv(ah.APP_HOME_ENV, raising=False)
20
21
22    # ---- default (env unset) is byte-identical to today ----
23
24
25    def test_app_home_defaults_to_cwd():
26        assert ah.app_home() == Path(".")
27
28
29    def test_default_config_path_unset_is_cwd_config_json():
30        # Pre-P0a this was the module constant Path("config.json").
31        assert ah.default_config_path() == Path("config.json")
32
33
34    def test_default_env_path_unset_is_cwd_dotenv():
35        assert ah.default_env_path() == Path(".env")
36
37
38    def test_resolve_output_dir_unset_is_verbatim_for_relative():
39        # env unset ? returned VERBATIM: no pathlib normalisation of ./ , separators, or
trailing slash
40        # (the pre-P0a code stored --output-dir as-is, and that string is echoed by
/api/config + reports).
41        for v in ("output", "custom", "./output", "out/clips", "output/"):
42            assert ah.resolve_output_dir(v) == v
43            assert ah.resolve_output_dir("") == "output" # empty ? the historical default
44
45
46    def test_env_file_to_load_unset_returns_none(monkeypatch, tmp_path):
47        # Even with a .env sitting in the CWD, the unset path must NOT prefer it ? bare
load_dotenv()
48        # walks up from the caller FILE, not the CWD, so None preserves that exact
behaviour.
49        monkeypatch.chdir(tmp_path)
50        (tmp_path / ".env").write_text("X=1\n", encoding="utf-8")
51        assert ah.env_file_to_load() is None
52
```

```

53
54 # ---- RALLY_APP_HOME set ? everything roots there ----
55
56
57 def test_app_home_honors_env(monkeypatch, tmp_path):
58     monkeypatch.setenv(ah.APP_HOME_ENV, str(tmp_path))
59     assert ah.app_home() == tmp_path
60
61
62 def test_app_home_expands_user(monkeypatch):
63     monkeypatch.setenv(ah.APP_HOME_ENV, "~/khelsutra-data")
64     assert ah.app_home() == Path.home() / "khelsutra-data"
65
66
67 def test_app_home_ignores_blank_env(monkeypatch):
68     monkeypatch.setenv(ah.APP_HOME_ENV, " ")
69     assert ah.app_home() == Path(".")
70
71
72 def test_config_and_env_paths_root_under_home(monkeypatch, tmp_path):
73     monkeypatch.setenv(ah.APP_HOME_ENV, str(tmp_path))
74     assert ah.default_config_path() == tmp_path / "config.json"
75     assert ah.default_env_path() == tmp_path / ".env"
76     assert ah.app_path("output", "x.db") == tmp_path / "output" / "x.db"
77
78
79 def test_resolve_output_dir_set(monkeypatch, tmp_path):
80     monkeypatch.setenv(ah.APP_HOME_ENV, str(tmp_path))
81     assert ah.resolve_output_dir("output") == str(tmp_path / "output")
82
83
84 def test_resolve_output_dir_absolute_passthrough(monkeypatch, tmp_path):
85     monkeypatch.setenv(ah.APP_HOME_ENV, str(tmp_path))
86     abs_dir = (
87         str(tmp_path / "elsewhere")
88         if os.path.isabs(str(tmp_path))
89         else os.path.abspath("elsewhere")
90     )
91     assert ah.resolve_output_dir(abs_dir) == abs_dir
92
93
94 def test_resolve_output_dir_empty_falls_back_to_home_output(monkeypatch, tmp_path):
95     monkeypatch.setenv(ah.APP_HOME_ENV, str(tmp_path))
96     assert ah.resolve_output_dir("") == str(tmp_path / "output")
97
98
99 # ---- default_os_app_home: the OS-conventional dir the launcher exports (NOT auto-
selected) ----
100
101
102 def test_default_os_app_home_windows(monkeypatch, tmp_path):
103     monkeypatch.setattr(ah.platform, "system", lambda: "Windows")
104     monkeypatch.setenv("APPDATA", str(tmp_path / "Roaming"))
105     assert ah.default_os_app_home() == tmp_path / "Roaming" / "Khelsutra"
106
107
108 def test_default_os_app_home_macos(monkeypatch):
109     monkeypatch.setattr(ah.platform, "system", lambda: "Darwin")
110     assert (
111         ah.default_os_app_home()
112         == Path.home() / "Library" / "Application Support" / "Khelsutra"
113     )
114
115
116 def test_default_os_app_home_linux_xdg(monkeypatch, tmp_path):

```

```

117 monkeypatch.setattr(ah.platform, "system", lambda: "Linux")
118 monkeypatch.setenv("XDG_DATA_HOME", str(tmp_path / "share"))
119 assert ah.default_os_app_home() == tmp_path / "share" / "Khelsutra"
120
121
122 def test_default_os_app_home_linux_default(monkeypatch):
123     monkeypatch.setattr(ah.platform, "system", lambda: "Linux")
124     monkeypatch.delenv("XDG_DATA_HOME", raising=False)
125     assert ah.default_os_app_home() == Path.home() / ".local" / "share" / "Khelsutra"
126
127
128 # ---- loader integration: load_config() roots on the app-home ----
129
130
131 def test_loader_reads_config_from_app_home(monkeypatch, tmp_path):
132     monkeypatch.setenv(ah.APP_HOME_ENV, str(tmp_path))
133     (tmp_path / "config.json").write_text('{"sport": "tennis"}', encoding="utf-8")
134     cfg = (
135         loader.load_config()
136     ) # no explicit path ? default_config_path() under the home
137     assert cfg.sport == "tennis"
138
139
140 def test_loader_no_config_in_home_uses_defaults(monkeypatch, tmp_path):
141     monkeypatch.setenv(ah.APP_HOME_ENV, str(tmp_path)) # empty dir, no config.json
142     cfg = loader.load_config()
143     # Defaults load cleanly (no crash, no file) ? same contract as a missing CWD
144     # config.json.
145     assert cfg is not None
146     assert hasattr(cfg, "indexing")
147
148
149 def test_save_default_config_writes_under_app_home(monkeypatch, tmp_path):
150     monkeypatch.setenv(ah.APP_HOME_ENV, str(tmp_path))
151     written = loader.save_default_config() # no explicit path
152     assert written == tmp_path / "config.json"
153     assert written.exists()
154
155
156 def test_explicit_path_still_overrides_app_home(monkeypatch, tmp_path):
157     monkeypatch.setenv(ah.APP_HOME_ENV, str(tmp_path / "home"))
158     explicit = tmp_path / "explicit.json"
159     explicit.write_text('{"sport": "pickleball"}', encoding="utf-8")
160     cfg = loader.load_config(str(explicit))
161     assert cfg.sport == "pickleball"
162
163
164 # ---- env_file_to_load: app-home .env preferred ONLY when RALLY_APP_HOME is set ----
165
166
167 def test_env_file_to_load_set_and_present(monkeypatch, tmp_path):
168     monkeypatch.setenv(ah.APP_HOME_ENV, str(tmp_path))
169     (tmp_path / ".env").write_text("X=1\n", encoding="utf-8")
170     assert ah.env_file_to_load() == tmp_path / ".env"
171
172
173 def test_env_file_to_load_set_but_absent_returns_none(monkeypatch, tmp_path):
174     monkeypatch.setenv(ah.APP_HOME_ENV, str(tmp_path)) # home pinned, but no .env in it
175     assert ah.env_file_to_load() is None
176
177
178 # ---- resolve_output_dir passthrough for absolute / Windows drive-anchored inputs ----
179
180
181 def test_resolve_output_dir_set_absolute_passthrough(monkeypatch, tmp_path):

```



```

181 monkeypatch.setenv(ah.APP_HOME_ENV, str(tmp_path))
182 abs_p = os.path.abspath(os.sep + "data" + os.sep + "reels")
183 assert ah.resolve_output_dir(abs_p) == abs_p
184
185
186 def test_resolve_output_dir_set_windows_drive_passthrough(monkeypatch, tmp_path):
187     monkeypatch.setenv(ah.APP_HOME_ENV, str(tmp_path))
188     if os.name == "nt":
189         # drive-anchored ("C:out" / "C:\\out") must NOT be re-rooted under the app home
190         assert ah.resolve_output_dir("C:out") == "C:out"
191         assert ah.resolve_output_dir("C:\\out") == "C:\\out"
192     else:
193         pytest.skip("Windows drive-letter semantics only meaningful on nt")
194
195
196 # ---- the package-relative static mount (CWD-independent) ----
197
198
199 def test_main_static_dir_is_package_relative():
200     import backend.main as m
201
202     expected = os.path.join(
203         os.path.dirname(os.path.abspath(m.__file__)), "api", "static"
204     )
205     assert m._STATIC_DIR == expected
206     assert os.path.isabs(m._STATIC_DIR)
207

```

20. assistant (2026-07-02T04:47:45.823Z)

Let me check how `create\_app` wires `global\_config`, and how main() resolves output_dir at startup.

21. user (2026-07-02T04:47:54.173Z)

```

98: Also sets the module-level ``global_config`` / ``db_instance`` for transitional
101: global global_config, db_instance
102: global_config = config
280:global_config: Optional[AppConfig] = None # now typed (Pydantic model)
1138: module-level ``global_config``/``db_instance`` (backward-compatible with existing
1157: config = global_config
1482:# backend.main.global_config, so behavior ? route paths, status codes, body-size limits,
1938: # global_config isn't initialized this early; load the typed config so the trim
1981: global global_config, db_instance
1982: global_config = load_app_config() # returns typed AppConfig
1984: global_config
1992: global_config.output.output_dir = resolve_output_dir(args.output_dir)
1993: os.makedirs(global_config.output.output_dir, exist_ok=True)
1996: global_config.output.output_dir, global_config.output.db_name
2039: _maybe_sweep_gemini_orphans(global_config)
2047: storage_cfg = getattr(global_config, "storage", None)
2064: local_video, global_config
2075: getattr(global_config, "indexing", None), "default_segmenter", "motion"
2101: segmenter = segmenter_cls(db_instance, global_config)
2127: config=global_config,
2165: global_config, args.host
2181: _get_executor().submit(_drain_due_jobs, global_config, db_instance)
2184: create_app(global_config, db_instance)
2190: # shared mutable state (`db_instance`, `global_config`) via `import backend.main` ? a
SEPARATE

```

22. user (2026-07-02T04:48:00.826Z)

```

1975 f"[APP] First run: seeded a truly-local default config at {seeded_path}

```

```

"
1976         f"(detector 'motion' ? offline, no API key or GPU needed). "
1977         f"Use the in-app setup to switch to Gemini."
1978     )
1979
1980     # Initialize Global Config and DB
1981     global global_config, db_instance
1982     global_config = load_app_config() # returns typed AppConfig
1983     _enforce_startup_or_exit(
1984         global_config
1985     ) # M5: fail fast on an incoherent ACTIVE profile
1986
1987     # The CLI --output-dir overrides the configured output directory. It now lives
1988     # on the typed model (OutputConfig.output_dir), so every consumer ? including
1989     # /api/compile ? reads the same value instead of a write-only runtime hack.
1990     # P0a: a RELATIVE output dir roots under the app-home (RALLY_APP_HOME); with the env
1991     # unset, resolve_output_dir("output") == "output" (CWD-relative) ? byte-identical to
before.
1992     global_config.output.output_dir = resolve_output_dir(args.output_dir)
1993     os.makedirs(global_config.output.output_dir, exist_ok=True)
1994
1995     db_path = os.path.join(
1996         global_config.output.output_dir, global_config.output.db_name
1997     )
1998
1999     # P0e: refuse a SECOND instance against the same DB BEFORE the destructive stale-job
sweep below.
2000     # The job worker is in-process, so a concurrent process opening this DB would sweep
THIS one's
2001     # in-flight jobs to 'failed'. The OS lock (held for the process lifetime; auto-
released on exit
2002     # or crash) is keyed to db_path, so a different --output-dir/DB still runs
concurrently.
2003     global _instance_lock
2004     try:

```

23. user (2026-07-02T04:48:15.142Z)

```

108
109     ## 2. SUBSYSTEMS
110
111     ### 2.0 Orchestration & config
112     - `@backend/main.py` ? Thin CLI + FastAPI bootstrap (orchestrator; modes via FLAGS).
Static UI mounts `/static` from `backend/api/static`. Logging is initialized with `basicConfig`
(INFO, timestamped format); backend modules use `logger = logging.getLogger(__name__)` (replaces
stray prints in hot paths like motion, stitcher, wasb_infer, config loader). User-facing CLI
summaries may still use `print` for direct output.
113     - `::app` FastAPI; `::db_instance`, `::global_config` (`Optional[AppConfig]`) module
globals set ONLY in `main()`. ? importing `app` for ASGI without `main()` -> both `None` ->
every endpoint NPES.
114     - **ASYNC (#16, PR-1):** `::process_video` `@POST /api/process` -> fast-fail 4xx
(path/exists/segmenter) -> `db.create_job` -> `_get_executor().submit(_drain_due_jobs)` -> **202
+ job_id**. `::drain_due_jobs` / `::run_claimed_job` (worker loop EXTRACTED to
`@backend/api/job_worker.py` (#234), re-exported on `backend.main`): claim each runnable job via
`db.claim_due_job` -> `_execute_processing` -> mark done/failed (a `JobWaiting` parks the job
via `set_job_waiting` for resume; EXECUTION_POLICY P3 self-scheduling). `::_execute_processing`
= the former sync body (hash -> validate -> add_video -> segment -> `finally:` always
`emit_indexer_report`, grafts `response["reports"]`; never raises from the report); a RAISING
segmenter now prune_after-restores pre-run rows before re-raising. `::get_job` `@GET
/api/jobs/{job_id}` -> `{status, progress, video_id, error, result, reports, ...}` ? job
`status` = EXECUTION state (`queued|running|done|failed`); the pipeline verdict lives in
`result["status"]`. `::_get_executor` lazy module-global `ThreadPoolExecutor(max_workers=1)`
(single box + Gemini serial-upload learning; tests monkeypatch it inline). `main()` runs
`db.fail_stale_jobs()` at startup (orphaned queued/running -> failed). `::compile_highlights`
`@POST /api/compile`. `::query_play_segments` `@POST /api/query`. `::get_config` `@GET

```

```

/api/config`. `::run_cli_diagnose_clip` `--debug-clip`: lazy `import cv2`, samples ±3s vs
`indexing.motion_threshold` (dual model/dict read). ? CLI `--video-path` mode still runs the
pipeline synchronously inline (separate code path, unchanged by #16).
115 - **UPLOAD/DOWNLOAD (PR-2, B2+B3):** the chunked-upload endpoints now live in
`@backend/api/uploads.py` (#228 ? an `APIRouter` registered via `app.include_router`;
`uploads.py::upload_cfg` resolves the live `backend.main.global_config` lazily so it reads the
startup config, and a lazy `import backend.main` avoids an import cycle) ?
`uploads.py::upload_init` `@POST /api/upload/init` (`{filename, total_size_bytes?}` -> ext
allowlist + size gate -> `{upload_id}`), `::upload_part` `@PUT /api/upload/{id}/part/{n}` (raw-
bytes body, STRICTLY sequential n, streams to `.partial-<id>` under `security.upload_dir`; per-
part/total cap breach -> 413 + upload ABORTED), `::upload_complete` `@POST .../complete`
(`{total_parts}` -> assemble -> never-overwrite rename -> `{path, video_id(MD5), size_bytes}` ?
client then POSTs /api/process with that path), `::upload_abort` `@DELETE /api/upload/{id}`. ?
upload state (`uploads.py::uploads` + `::uploads_lock`) is IN-PROCESS ? restart aborts
partials (client re-inits); per-user partitioning lands with PR-3. `main.py::download_highlight`
`@GET /api/highlights/{video_id}/{filename}` (download was NOT moved ? still in main.py) ? safe-
name + `resolve_under_root(output_dir)` then FileResponse stream (the old /api/compile alert
path was browser-unreachable). Config: `SecurityConfig.{upload_dir='output/uploads',
max_upload_mb=4096, max_part_mb=95, allowed_upload_extensions=[mp4,mov]}` (95MB: free-tier edge
proxies cap bodies ~100MB).
116 - **STUDIO / STORAGE endpoints (P1/P2/P3/P4/P6/P8, 2026-07-01):** `::capabilities`
`@GET /api/capabilities` ? honest box probe; now includes a **`storage` block**
(`storage_capabilities` in `@backend/storage/capabilities.py`: local/s3/gcs/gdrive
`{id,kind,label,usable,free_gb}`, secret-free) alongside segmenters/ffmpeg/gpu/deployment.
`::list_videos` `@GET /api/videos` (the "my videos" list, P6). `::get_source` `@GET
/api/source/{id}` ? Range-capable FileResponse serving a **frame-identical annotation proxy**
(D4), source fallback, background-warmed. `::post_annotations` `@POST /api/annotations` ? writes
the verbatim annotator CSV to `output/labels/<video_id>.rallies.csv` (P8). `::create_asset`
`@POST /api/assets` (**P4**, `{uploaded_path?|source_uri?, medium_uri, sport?}`) ? stores a
video into a user-chosen medium (MD5-keyed `storage.put` for an upload?remote medium; register-
in-place for local; at-rest URI = no copy), registers it (video_id=MD5) so it shows in
`/api/videos`; **self-host posture only** (403 when `security.allowed_video_root` set); edge-
auth mirrors `/api/process`. Cost ledger: `::get_job_cost` `@GET /api/jobs/{id}/cost`,
`::get_video_cost` `@GET /api/videos/{id}/cost`. **Free-space guard (P2):**
`@backend/utils/freespace.py::require_free_bytes` -> HTTP **507** in `uploads.py::upload_init` +
the `/api/process` remote fetch-to-scratch; the WSL frame-extraction guard now **BLOCKS**
(`detectors/base.py::wsl_frame_stage_shortfall`, ~15x source).
117 - `::load_config` ? ? legacy thin wrapper returning a **dict**
(`load_app_config(path).model_dump(...)`); kept for transition ? new code imports
`load_app_config`/`AppConfig` directly.
`::ProcessRequest`/`::QueryRequest`/`::CompileRequest`/`::AnnotationsRequest`/`::AssetRequest`
pydantic bodies.
118 - `main()` sets `global_config.output.output_dir = args.output_dir` (typed field on
`OutputConfig`); `db_path = os.path.join(global_config.output.output_dir, output.db_name)`.
`compile_highlights` reads the SAME `global_config.output.output_dir`, so `--output-dir` is
honored end-to-end (the old write-only `_runtime_overrides` hack is gone).
119 - ? `--segmenter` argparse `choices` frozen at build time from
`segmenter_registry.list_available()` ? a lazily-registered segmenter wouldn't appear. CLI
default_segmenter fallback `motion`.
120 - `@scripts/run_process_and_stitch.py` ? one-shot demo; ? hardcoded paths, interactive
`input()` (blocks automation), skips validation. Config via `backend.config.load_config`
(typed); segmenter, padding, and db path all read from the model (no literal fallbacks).
121 - `@run_eval.py::main` ? score one video's DB preds vs GT CSV.
`--stage{candidates,final,both}`. Exit 2=arg/IO. ? pure scorer; errors if video never processed.
`::default_db_path` resolves via `backend.config.load_config`
(`output.output_dir`/`output.db_name`). `get_file_md5` is an alias for `compute_video_id`.
122 - `@scripts/run_phase3_eval.py::main` ? walk manifest, segment-if-needed, score,
aggregate. `load_dotenv()` at import. ? DB key = file MD5, NOT manifest's `video_id` (kept as
label); config via `backend.config.load_config`; `segmenter_cls(db, cfg)` receives the typed
`AppConfig` (same object the live pipeline feeds segmenters).
123 - `@run_regression.py::main` ? `regress_with_provider` vs baseline. Exit codes semantic:
**0=ok, 1=REGRESSION (CI gate), 2=missing DB**. ? `--update-baseline` runs AFTER compare
(snapshots even a regressing run); `::default_db_path` resolves via
`backend.config.load_config`.

```

24. user (2026-07-02T04:48:27.231Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-upload-dir/backend/api/uploads.py has been updated successfully. (file state is current in your context ? no need to Read it back)

25. user (2026-07-02T04:48:34.499Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-upload-dir/backend/api/uploads.py has been updated successfully. (file state is current in your context ? no need to Read it back)

26. user (2026-07-02T04:48:42.778Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-upload-dir/backend/api/uploads.py has been updated successfully. (file state is current in your context ? no need to Read it back)

27. assistant (2026-07-02T04:48:46.674Z)

Now update the config model comment to document the app-home rooting.

28. user (2026-07-02T04:48:51.071Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-upload-dir/backend/config/models.py has been updated successfully. (file state is current in your context ? no need to Read it back)

29. user (2026-07-02T04:49:01.521Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-upload-dir/docs/CODE_MAP.md has been updated successfully. (file state is current in your context ? no need to Read it back)

30. assistant (2026-07-02T04:49:27.720Z)

Now add the tests. I'll extend `tests/test\_upload\_download.py` with a P0a app-home section.

31. user (2026-07-02T04:49:31.760Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-upload-dir/tests/test_upload_download.py has been updated successfully. (file state is current in your context ? no need to Read it back)

32. user (2026-07-02T04:49:49.698Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-upload-dir/tests/test_upload_download.py has been updated successfully. (file state is current in your context ? no need to Read it back)

33. assistant (2026-07-02T04:49:56.749Z)

Now run the targeted tests and lint.

34. user (2026-07-02T04:50:06.454Z)

..... [100%]
46 passed in 4.99s

35. user (2026-07-02T04:50:16.676Z)

```
Exit code 2
All checks passed!
3 files already formatted
mypy.ini
pyproject.toml
ruff.toml
```

36. assistant (2026-07-02T04:50:22.099Z)

mypy config exists ? run mypy on touched modules.

37. user (2026-07-02T04:51:34.701Z)

```
backend\annotations\base.py:88: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
backend\sports\__init__.py:9: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
backend\pipeline\validators\base.py:44: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
backend\providers\__init__.py:9: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 2 source files
```

38. assistant (2026-07-02T04:51:40.844Z)

All green. Quick link-check since I touched CODE_MAP.md, then commit.

39. user (2026-07-02T04:51:44.659Z)

```
[check-md-links] clean ? no broken internal links in live docs.
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
8b16cf4 docs: reflect today's storage endpoints + the corpus/user-video boundary (#411)
aa915a7 deploy/gcp: centralize project + bucket names in one sourced buckets.env (#410)
52d97b5 P4: POST /api/assets ? store a video into the user's chosen medium (#409)
```

40. user (2026-07-02T04:51:51.519Z)

```
fix/upload-dir-app-home
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
M backend/api/uploads.py
M backend/config/models.py
M docs/CODE_MAP.md
M tests/test_upload_download.py
```

41. user (2026-07-02T04:52:01.080Z)

```
[fix/upload-dir-app-home a8e191e] api: root a relative security.upload_dir under the app home
(P0a parity with output_dir)
4 files changed, 80 insertions(+), 5 deletions(-)
a8e191e api: root a relative security.upload_dir under the app home (P0a parity with output_dir)
```

42. user (2026-07-02T04:52:09.838Z)

```
1      <!-- Describe the change and link the issue it closes. -->
2
3      ## Summary
4
5
6
7      ## Checks
8      - [ ] `python run_all_tests.py` green; `ruff check .` + `mypy` clean; coverage not
lowered.
9      - [ ] Internal markdown links valid (`python -m tools.check_md_links`).
```

```
10
11     ## Launch / "bit-flip" PRs only
12     A PR that flips a default / "launch bit" so a tracked project reaches terminal state
MUST also run
13     the `docs/DOC_STATUS.md` §2 archival due-diligence on that project's docs, **in this
same PR**:
14     - [ ] Verified each doc's claims against the code (tagged load-bearing facts
`[verified]`).
15     - [ ] Flipped the doc Status ? DELIVERED/DONE/REJECTED + a completion note (what
shipped, PR, paths).
16     - [ ] Updated all inbound references (`docs/README.md`, `CLAUDE.md`,
`docs/NEXT_STEPS.md`, cross-doc links).
17     - [ ] Kept the ADR forward/back lineage.
18     - [ ] `git mv` the terminal docs to `docs/archives/?` and recorded the move in
`DOC_STATUS.md` §3.
19     - [ ] Fixed any internal-link breakage from the move (`link-check` CI job green).
20
21     _(Not a launch/bit-flip PR? Delete this section.)_
22
```

43. user (2026-07-02T04:52:22.661Z)

```
remote:
remote: Create a pull request for 'fix/upload-dir-app-home' on GitHub by visiting:
remote:     https://github.com/Khelsutra/badminton-highlight-indexer/pull/new/fix/upload-dir-
app-home
remote:
branch 'fix/upload-dir-app-home' set up to track 'origin/fix/upload-dir-app-home'.
To https://github.com/Khelsutra/badminton-highlight-indexer.git
* [new branch]      fix/upload-dir-app-home -> fix/upload-dir-app-home
```

44. user (2026-07-02T04:52:45.708Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/pull/415>

45. assistant (2026-07-02T04:52:55.748Z)

PR #415 is open. Final verification of branch isolation was done pre-push (only my commit on top of origin/master).

46. user (2026-07-02T04:53:09.373Z)

Structured output provided successfully